## AI Solution

Modern healthcare systems operate as critical, high-volume public service industries where resource constraints, bed allocation bottlenecks, and clinician administrative burnout frequently lead to operational paralysis. The proposed system, Triage AI, directly aligns with the project theme, "An AI Solution for Industries," by deploying Fourth Industrial Revolution (4IR) technologies specifically supervised machine learning, deep learning telemetry analysis, time-series forecasting, and natural language processing to transition healthcare operations from reactive crisis management into proactive, data-driven workflows. By automating clinical documentation dictation, forecasting patient surge volumes, and identifying patients at high risk of early 30-day readmission prior to discharge, Triage AI provides public and regional hospital administrations with cost-efficient, scalable operational intelligence that maximizes bed utilization and enhances clinical service delivery.

## Problem Definition:

Public healthcare facilities and regional hospitals consistently face severe operational bottlenecks characterized by overcrowded casualty wards, prolonged emergency wait times, and ICU bed shortages. Hospital administrations lack predictive decision-support systems to forecast patient surge volumes, resulting in severe clinical understaffing during peak demand periods. Concurrently, medical staff spend up to 35% of their working hours manually completing administrative charts and electronic records, diverting vital attention from acute clinical triage. Furthermore, without automated predictive screening at discharge, vulnerable chronic patients are frequently released without structured follow-up, driving high rates of preventable 30-day hospital readmissions that strain already scarce clinical resources.

## Theme Relevance and Municipal Benefit

This challenge exemplifies an industrial inefficiency where 4IR AI solutions can optimize workflow capacity. Solving this problem directly benefits the local municipality and regional healthcare districts by lowering operational expenditure per patient, reducing casualty congestion, and ensuring intensive care beds remain available for local emergencies. Automated predictive triage strengthens public health resilience, improves patient outcomes across local communities, and maximizes municipal healthcare infrastructure efficiency.

## Business Objectives

### Business Background

Hospitals in developing economies operate under chronic budgetary constraints and high patient-to-clinician ratios. In South Africa and similar public health environments, tertiary and regional hospitals bear the dual burden of infectious diseases and rising chronic conditions (such as diabetes, cardiovascular disease, and hypertension). When administrative and clinical systems operate in siloes, hospital management cannot anticipate when an influx of admissions will occur or identify which discharged patients are likely to deteriorate. Implementing an integrated AI operational platform provides the analytical layer necessary to optimize workforce schedules, bed availability, and post-discharge interventions.

### Business Objectives

The strategic objectives of the Triage AI system are:

**1. Reduce 30-Day Preventable Readmissions:** Identify high-risk patients before discharge to enable targeted outpatient interventions and reduce readmission rates by 15% within the first operational year.

**2. Optimize Clinical Shift & Bed Planning:** Provide 14-week look-ahead forecasts of emergency department arrivals to decrease emergency waiting room times by 20%.

**3. Alleviate Administrative Clinician Burnout:** Implement speech-to-text dictation and medical entity parsing to reduce daily clinician charting time by at least 25%.

**4. Improve Critical Care Responsiveness:** Continuously evaluate physiological ICU vitals to alert nursing staff to acute patient deterioration prior to clinical shock.

### Business Success Criteria

#### Quantitative Criteria

• Achievement of a Machine Learning Receiver Operating Characteristic Area Under the Curve (ROC-AUC) score exceeding 0.65 on benchmark patient cohorts.

• A Time-Series Mean Absolute Error (MAE) under 15 patients per week on admission volume projections.

• Reduction of average non-acute emergency waiting times from 4 hours to under 2.5 hours.

**Qualitative Criteria**

• Positive clinical adoption and usability feedback from medical personnel using the Clinical Assistant Softbot for operational and bed status queries.

• Compliance with clinical governance and data privacy legislation.

**Requirements Analysis**

**Functional Requirements:**

1. The system must ingest structured Electronic Health Records (EHR) and output a calibrated 30-day readmission risk score prior to patient discharge.
2. The system must aggregate historical admissions and produce automated seasonal volume forecasts with 95% confidence intervals.
3. The system must transcribe clinical voice dictation and extract medical entities (symptoms, medications, dosages).
4. The system must provide a conversational softbot to answer staff operational queries (bed capacity, triage categories, emergency escalation codes).

**Non-Functional Requirements:**

1. **Performance:** Inference response time for risk scoring must be under 2 seconds per patient record.
2. **Security & Compliance:** System design must comply with data protection regulations (such as POPIA) through pseudonymization of patient identifiers.
3. **Availability:** The operational softbot and alerting pipeline must maintain 99.5% uptime within the hospital intranet.

**Constraints and Risk Management**

| Dimension | Description | Mitigation Strategy |
|---|---|---|
| Constraint: Data Privacy (POPIA) | Access to live identifiable South African patient records is restricted by privacy law. | Model development and validation use de-identified benchmark datasets (UCI 130-US Hospitals) with synthetic local validation testbeds. |

| Dimension | Description | Mitigation Strategy |
|---|---|---|
| Risk: Class Imbalance in Clinical Data | In hospital readmission cohorts, only ~9% to 11% of patients represent positive early readmission instances. | Utilize balanced class weighting (class_weight='balanced') and focus on Recall and ROC-AUC rather than raw classification accuracy. |
| Risk: Staff Adoption Resistance | Clinical staff may view software interaction as an additional administrative burden. | Provide low-friction interfaces (voice dictation and natural language conversational softbot queries) rather than complex manual forms. |
| Risk: Model Concept Drift | Seasonal disease outbreaks or demographic shifts can degrade predictive accuracy over time. | Schedule quarterly model retraining runs and monitor performance against baseline test metrics. |

**Initial Assessment of Tools and Techniques**

**Programming Environment & Execution:** Python 3.12 within VS Code, managed through a structured Git repository and GitHub Projects Kanban board.

**Data Processing & Analysis:** Pandas and NumPy for matrix operations, categorical one-hot encoding, and missing value imputation.

**Machine Learning and Time Series**

**Scikit-Learn:** Selected for Random Forest classification, StandardScaler normalization, and stratified test partitioning.

**Statsmodels:** Selected for SARIMAX (Seasonal Autoregressive Integrated Moving Average) modeling to capture underlying trends and seasonal hospital surges.

**Deep Learning:** PyTorch: Selected for constructing Multi-Layer Perceptrons to evaluate real-time multivariate patient vital sign streams.

**NLP & Speech Synthesis:** spaCy and regular expression pattern matchers for clinical entity recognition; pyttsx3 for text-to-speech operational dispatch alerts.